Philosophy of Space and Time

Lecture 10: *Special Relativity and the Relativity of Simultaneity*

Chris Smeenk

Department of Philosophy, UCLA · April 30, 2026

Today's Plan

Where We Left Off

On Tuesday we argued that:

- Light signals are the simplest tool for probing spacetime structure
- The speed of light is *source-independent* (binary star argument)
- This is incompatible with Galilean velocity addition
- Einstein's resolution: drop absolute simultaneity; replace simultaneity slices with **light cones**

Today we work out *what it means* to give up absolute simultaneity — this is the primary conceptual innovation of special relativity.

Einstein's Two Postulates

Einstein (1905), "On the Electrodynamics of Moving Bodies," builds the theory on just two postulates:

Postulate 1: Principle of Relativity

The laws of physics take the same form in all inertial frames. No experiment can single out a preferred frame.

Postulate 2: Constancy of the Speed of Light

Light propagates with a definite speed c, independent of the motion of the source.

?? *Widely held before Einstein. The revolution is showing that they are in fact compatible, which requires giving up absolute simultaneity.*

What Does Simultaneity Mean?

A Question We Usually Take for Granted

When we say that two events happen "at the same time," what *exactly* do we mean?

- For events at the *same place*, this is easy: we can tell directly
- For events at *different places*, we cannot simply "look"—we need a procedure
- Prior to Einstein, physicists assumed simultaneity was simply *given*—a basic, objective feature of the world

?? *Einstein's move: if we cannot simply "look," we had better give an explicit operational procedure for determining whether two distant events are simultaneous.*

A Naïve Procedure

First attempt: send a fast messenger.

- An event q occurs somewhere at a distance from me
- Send a runner to the location of q; have them note the time on their watch
- Compare to my watch

?? *This fails: the runner's watch may go out of sync, or they may travel at uneven speed. To make the procedure rigorous, we need a signal whose propagation we understand precisely—one that does not depend on the vagaries of a material messenger.*

Einstein's Procedure: The Radar Method

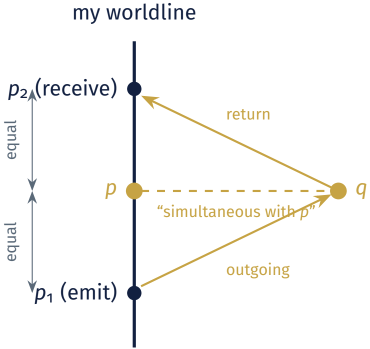
Einstein's approach (1905), followed by Geroch: a single observer with a clock and the ability to send and receive light signals.

- Along my worldline, let p be some event on my clock
- Consider a distant event q. Is q simultaneous with p?
- **Send** a light signal toward q at an earlier event p_1 on my worldline
- The signal reflects off q and **returns** to me at a later event p_2
- *Declare* q simultaneous with p iff p is the midpoint of the interval from p_1 to p_2 on my clock

Why This Works

By Postulate 2, light travels at speed c regardless of state of motion of emitter. So it is plausible to take q as happening halfway between emission and reception.

The Einstein Simultaneity Criterion



Event q is simultaneous with p (in my frame) iff p is the midpoint—on my clock—of a round-trip light signal to q and back.

What the Procedure Does

This definition is not arbitrary—it is forced by the structure of our situation:

- Uses only the observer's clock and light signals
- Clear procedure for carrying it out
- Only the observer's proper time, not any "universal" time, is used

?? *The procedure is frame-relative. A different inertial observer, moving relative to me, carrying out the same round-trip procedure using their clock will reach a different conclusion about whether q is simultaneous with p.*

Illustration

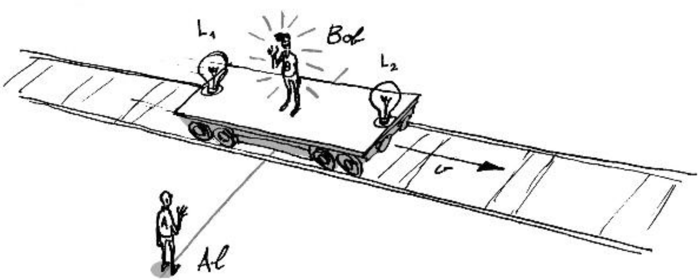
Example of Relativity of Simultaneity

Consider two inertial observers in relative motion, each carrying out the procedure:

- Al stands on the platform, at rest relative to the tracks
- Bob rides a long rail car, standing at its *middle*
- Two lights L_1 and L_2, fixed at the two ends of the rail car, flash once each
- The two flashes are timed so that they reach Bob at the moment he passes Al

The question: did the two lights flash *at the same time*?

The Setup



Bob stands in the middle of the rail car; lights L_1 (rear) and L_2 (front) flash once each. Both flashes reach Bob as he passes Al. The rail car moves at velocity v relative to the platform.

Al's Verdict

Al reasons as follows:

- The two flashes both reach Bob at the same moment
- By Postulate 2, light from each flash travels at speed c
- Bob is moving away from L_1, so the flash has to cover a distance $>$ half the length of the rail car
- Bob is moving towards the L_2, so the flash covers a distance $<$ half the length of the rail car
- So L_1 had to flash earlier than L_2

Al says:

The two lights did not flash simultaneously.

(Note: I have corrected this slide.)

Bob's Verdict

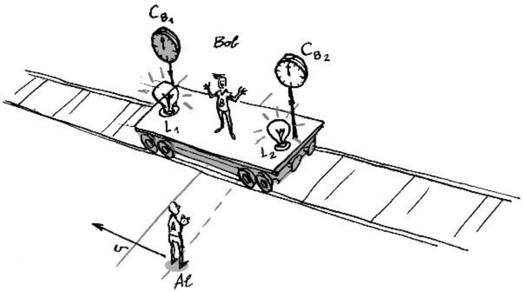
From Bob's perspective, the rail car is at rest; the flashes reach him at the same moment.

- Bob is at the middle of the rail car, equidistant from L_1 and L_2
- By Postulate 2, light travels at speed c in both directions in Bob's frame
- The flashes arrive together; distances are equal; speed is equal in each direction
- Therefore the flashes were emitted at the same moment

Bob says:

The two lights flashed simultaneously.

Bob's Full Picture



Bob adds clocks C_{B_1} and C_{B_2} next to each light, synchronized in his frame. In Bob's description, the two clocks read the *same* time when the two lights flash.

Al's Verdict

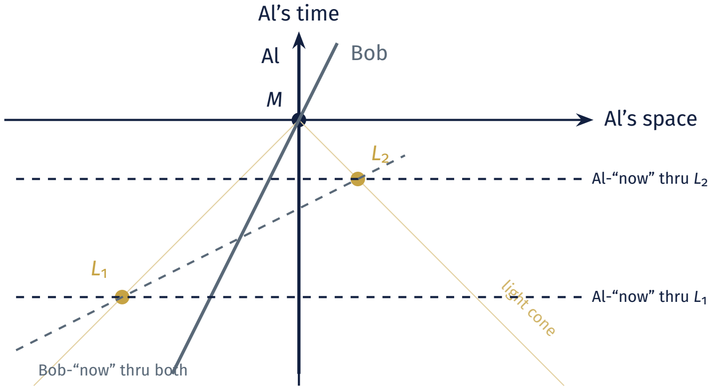
Al says (about Bob's clocks):

C_{B_1} and C_{B_2} are not synchronized correctly, they assign different times to simultaneous events.

Bob and Al have applied the *same* simultaneity criterion. They reach *different* conclusions about timing of the two flashes. Neither has made a mistake.

This is the **relativity of simultaneity**.

Why Both Are Right: The Spacetime Picture



In Al's frame, L_1 and L_2 lie on *different* horizontal "nows"—Al sees L_1 earlier. Both lie on a *single* Bob-"now"—for Bob, they are simultaneous.

The Logic of the Argument

The argument uses only two ingredients:

1. **Einstein's simultaneity criterion**: event q is simultaneous with event p on my worldline iff p is the midpoint (on my clock) of a round-trip light signal to q
2. **The constancy of c**: light travels at c in *any* inertial frame, independent of source or observer

- Al applies the criterion in his frame: gets one answer
- Bob applies the *same* criterion in his frame: gets a different answer
- Both must be right, because each has followed the operational procedure

?? *The result: "simultaneity" does not pick out a single, frame-independent relation between events. It is frame-relative.*

Where the Classical Picture Goes Wrong

In the Galilean picture, simultaneity is *absolute*: there is a unique way to slice spacetime into “layers of now.”

- All inertial observers agree on this slicing
- The slicing is part of the spacetime structure

In the Einsteinian picture:

- Each inertial observer has *their own* slicing into layers of “now”
- Different observers’ slicings disagree for distant events
- There is no preferred slicing—no objective fact about which slicing is “right”
- Simultaneity is part of the observer’s *description*, not part of the spacetime itself

Maudlin's Commentary

Maudlin emphasizes (Ch. 4): the relativity of simultaneity is not a paradox, a trick, or a problem of perception.

- It is not that Al and Bob are "fooled" by their motion
- It is not that one of them has made a measurement error
- It is not that they are using different definitions of "simultaneous"
- They are using the *same* definition—and getting different answers

The Lesson

What is simultaneous depends on your state of motion / which inertial frame you occupy. Different frames cut spacetime into different "layers of now."

Break

Relativity of Other Quantities

Further consequences of Simultaneity

Once simultaneity becomes frame-relative, other classical quantities do too:

- **Length** of a moving object: measure by locating both ends *at the same time*. But "same time" depends on the frame.
- **Duration** of a process happening at different locations: need to synchronize clocks at the two locations. But what counts as "synchronized" depends on the frame.
- **Simultaneity of distant events**: already shown relative above.

?? *Everything that presupposes a notion of "same time at different places" becomes frame-relative. This is why length and time intervals get reshuffled in special relativity—they all rest on simultaneity.*

A Geometric Picture: Slicing Spacetime

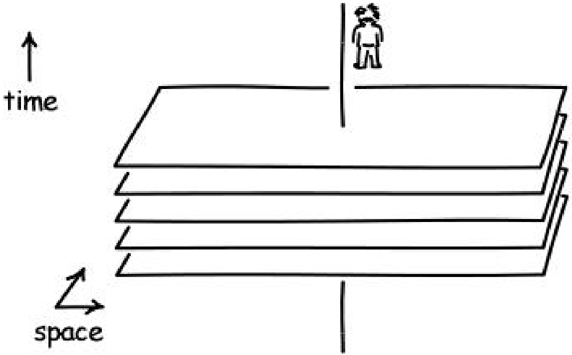
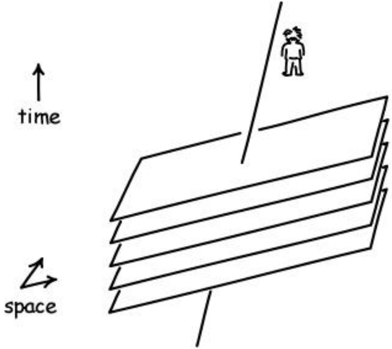


Image from John Norton's Einstein for Everyone

An inertial observer's "layers of now" can be pictured as a stack of parallel planes cutting through spacetime:

- Each plane is the set of events simultaneous (in that frame) with one moment on the observer's worldline
- The observer's worldline is perpendicular to their own simultaneity planes
- Time flows "upward" along the worldline; space extends outward along each plane

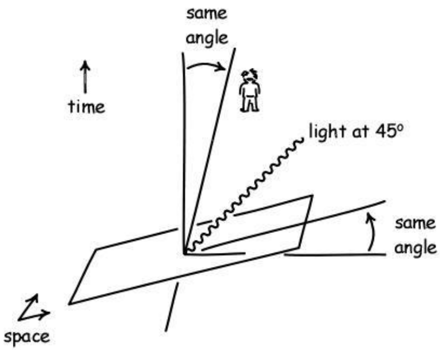
Different Observers, Different Slicings



An observer moving at velocity *v* relative to the first has a *different* stack of simultaneity planes:

- Their worldline tilts away from the vertical
- Their simultaneity planes tilt away from the horizontal—by the same angle
- The two slicings disagree about which distant events are simultaneous
- Neither slicing is privileged

The Light Cone as the Reference



The constancy of c constrains how the slicings tilt:

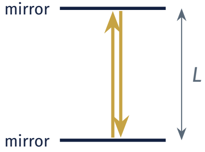
- Light propagates at 45° in any inertial frame
- When the worldline tilts toward the light cone, the simultaneity planes tilt toward it by the *same* angle
- The light cone is the one structure both observers agree on

Time Dilation

The Light Clock

A simple clock: a photon bouncing between two parallel mirrors separated by distance L.

- One "tick" $=$ a round trip
- In the clock's rest frame: $\Delta t = 2L/c$
- Simple, deterministic, and tied directly to the speed of light

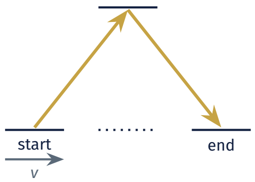


Now: what does an observer moving past this clock see?

The Light Clock in Motion

An observer in frame S sees the clock moving with speed v to the right. The photon's path is no longer vertical—it traces a zig-zag.

- Path length is *longer* than $2L$
- But by the second postulate, the photon still travels at c
- So the round-trip time *must* be longer than $2L/c$
- This is **time dilation**—directly forced by the constancy of c



Reciprocity

Crucially: the relation is *symmetric*. An observer co-moving with the clock can run an identical argument about *my* clock.

- I say: “her clock is running slow”
- She says: “*his* clock is running slow”
- Both claims are correct in their respective frames

?? *This is not a contradiction. Each is comparing the moving clock to a network of synchronized clocks at rest in their own frame. The two networks slice spacetime differently—and the slicing is what produces the asymmetry.*

What Time Dilation Is Not

Common misconceptions:

- *Not* a perceptual or signal-delay effect: it is what is measured *after* correcting for light-travel times
- *Not* a property of one particular type of clock: the same factor applies to atomic clocks, biological clocks, decay rates, all physical processes
- *Not* a violation of the principle of relativity: each frame is equally good, and each finds the other's clock slow

Maudlin's framing (Ch. 4): time dilation is a *geometric fact* about Minkowski spacetime, not a physical mechanism acting on clocks.

Length Contraction

Measuring a Moving Rod

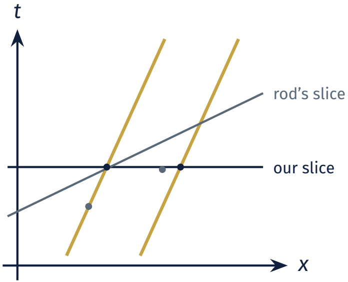
How do we measure the length of a rod moving past us?

- Mark the position of the front and back *at the same time* in our frame
- Then measure the distance between those marks
- In the rod's rest frame: length is L_0 (**proper length**)
- In our frame: what do we get?

?? *The answer depends on how we assess whether the marks were simultaneous. Since simultaneity is frame-relative, length will be too.*

The Geometry of Length Contraction

- The rod's worldlines (front and back) are two parallel lines in spacetime
- Its rest length: distance between them along a slice of *its* simultaneity
- Our measured length: distance between them along a slice of *our* simultaneity
- These slices intersect the worldlines at *different* pairs of events
- Result: $L = L_0/\gamma$



Reciprocity Again

Just as with time dilation:

- I find that her rod is shorter than its rest length
- She finds that *my* rod is shorter than its rest length
- Both are correct in their respective frames

Length Contraction

A rod of proper length L_0 moving with speed v has length
$L = L_0\sqrt{1 - v^2/c^2} = L_0/\gamma$ in our frame.

Like time dilation, this is a *geometric* fact about how different observers slice spacetime—not a physical compression of the rod.

What Remains Invariant?

Not everything becomes relative. Some quantities are the same for *all* inertial observers:

- The **speed of light** c (by postulate)
- The **light-cone structure** at each event
- The **spacetime interval** between two events (next week)
- **Causal order**: which events can influence which
- **Proper time** along a worldline (what a clock carried by an observer actually reads)

The Guiding Idea

Special relativity does not abolish objectivity—it re-locates it. What is objective is the spacetime interval (and derived notions), not space and time separately.

A Preview: The Interval

The quantity that all inertial observers agree on is the **spacetime interval**:

$$\Delta s^2 = -(c\,\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2$$

- Individual terms change from one frame to another (time dilation, length contraction)
- But the *combination* above is the same for all inertial observers
- This is analogous to how individual coordinates (x, y) change under rotation, but distance $\sqrt{x^2 + y^2}$ is invariant
- The minus sign in front of the time term is what makes spacetime *Minkowskian* rather than Euclidean

We will develop this in detail after the exam (Geroch Chs. 5–6; Maudlin Ch. 4).

Summary

1. **Einstein's postulates**: principle of relativity + constancy of c. Together they force a revision of simultaneity.
2. **Operationalizing simultaneity**: event q is simultaneous with event p on my worldline iff p is the midpoint of a round-trip light signal to q and back—as timed on my own clock.
3. **The train thought experiment**: Al and Bob, applying the same simultaneity criterion from different inertial frames, disagree about whether two distant events are simultaneous. Both are right.
4. **The relativity of simultaneity**: is not a perceptual illusion or a paradox—it is a consequence of the constancy of c applied consistently.
5. **Other classical quantities become frame-relative**: length contraction, time-dilation illustrate this.
6. **Looking ahead**: But there is an invariant geometric quantity—the spacetime interval.

Exam Logistics (Tuesday, May 5)

- Three questions now posted on the website; you will write on *one* (your choice)
- Usual place and time, 110 minutes total

For Thursday, May 7 (After the Exam)

- **Read:** Maudlin, Ch. 4 (continued); Geroch, Ch. 5.
- **Think about:**
  - Given Bob's and Al's disagreement about simultaneity, can they agree on *anything*?
  - What features of spacetime are still observer-independent?

Next week we develop:

- The **spacetime interval** and its invariance
- **Time dilation** and **length contraction** as consequences
- The **twins "paradox"**
- Geroch's operational construction of the interval from clocks and light signals

Overflow: The Magnet and Conductor

Einstein's 1905 paper opens with a puzzle from electromagnetism:

- Case 1: a magnet moves past a stationary conductor. The changing magnetic field induces an electric field in the conductor, producing a current.
- Case 2: the conductor moves past a stationary magnet. The charges in the moving conductor experience a force from the magnetic field, producing a current.
- *The current is the same in both cases*, but pre-relativistic theory treats them as fundamentally different phenomena

Einstein: observationally, only the *relative* motion matters. So the theory should not distinguish the cases. This suggests the principle of relativity applies to electromagnetism too—not just mechanics.

Overflow: Why the Train Example Is Canonical

Einstein introduced the train example in his 1917 popular book. It has become the standard illustration because:

- It uses only everyday objects and familiar motion
- It isolates the key assumption (constancy of c) from the conclusion (relative simultaneity)
- It requires no calculation—just careful reasoning about signal arrivals
- It shows clearly that the relativity of simultaneity is a *logical* consequence of the postulates, not a weird empirical fact

Maudlin's treatment in Ch. 4 follows this same argumentative structure.

Overflow: Is There a “True” Simultaneity?

A tempting reaction: “Surely *one* of Al and Bob must be right about the flashes of light? Either they really were simultaneous or they really weren’t.”

- This reaction presupposes that there is an objective, frame-independent fact of the matter
- But the argument shows: no operationally accessible procedure can identify one such fact
- To posit such a fact would be to reintroduce an unobservable structure—a preferred frame
- That is exactly what Einstein’s postulate 1 (principle of relativity) rules out

This is structurally the same move as the Galilean rejection of absolute velocity: unobservable structure should not be postulated.

Overflow: Simultaneity and Conventionality

A subtler philosophical question: is Einstein's simultaneity *conventional*?

- Reichenbach and others argued: the midpoint procedure assumes light travels at the same speed in both directions
- But we can only test this *by assuming* a simultaneity convention—circularity
- Conclusion (Reichenbach): Einstein simultaneity is a convention, not a discovery
- Response (Malament, others): within a given inertial frame, Einstein simultaneity is the unique relation satisfying natural symmetry conditions

This is one of the subtler issues in the philosophy of special relativity; we may return to it.

Questions?

chris.smeenk@gmail.com